Introduction

The task required me to treat the data as unlabelled and to employ unsupervised learning methods to determine the classes of the audio. Naturally, I turned to K Means Clustering as it relies on calculating similarities between data points by using Euclidean distance.

Data Preparation

Even though the assignment told me to treat the data as unlabelled, labels were still present as I only had to concern myself with a subset of the dataset (i.e. the animal classes). I filtered the animal audios out and essentially converted each audio file into an array of machine-readable numbers using Librosa.

Feature Extraction

There were many features relating to audio but I chose the minimum number of features that could theoretically still identify one type of audio from another. Adding too many features could potentially hinder the clustering process as some features may not contain meaningful clustering power (i.e. it is not a strong independent determinant for one audio to be distinct from another).

Firstly, I chose Root Mean Square (RMS) Energy because it gives a good indication of how loud the sound is. For instance, a dog's bark is usually louder than a cat's meow.

Secondly, I chose Zero-Crossing Rate (ZCR) to measure how fast the audio changes (or in technical terms, how often it crosses the horizontal axis). A crow's waveform is much likely to change rapidly compared to a cow.

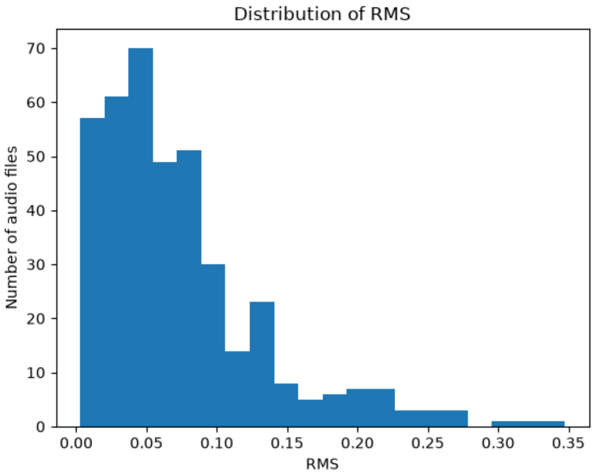
Thirdly, I chose Spectral Centroid because it gives a good measure of the level of frequency most of the audio's energy is at. A crow's audio energy is likely to be clustered at higher frequencies while the cow's audio energy is likely to be clustered at lower frequencies.

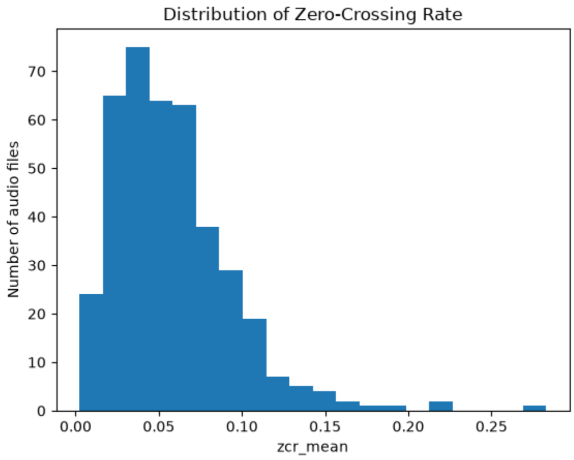
Lastly, I chose Spectral Bandwidth because it illustrates the spread of frequencies in an audio clip. A dog's barking is likely to span across multiple frequencies as opposed to a cow's moo.

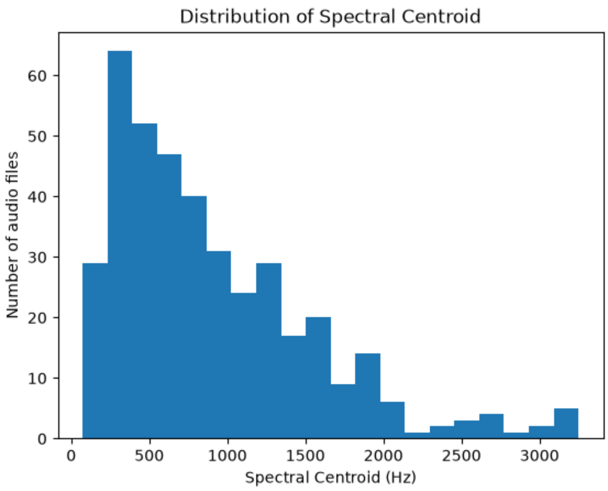
For each audio clip, I chose to take the mean of all features across all samples in the audio clip to get a rough sensing of how the audio performed across all features across the 5 seconds of the clip.

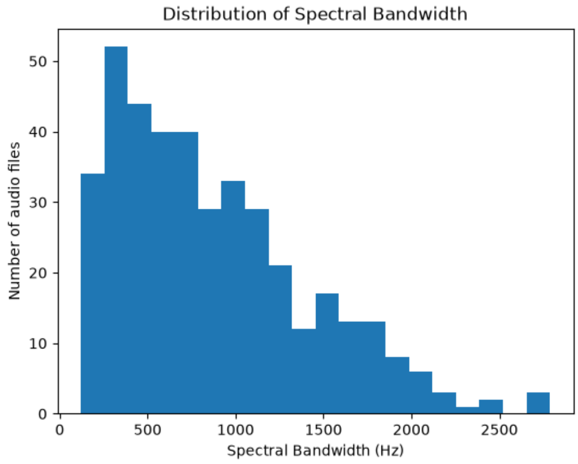
Exploratory data analysis

Now that I had locked down my features, I attempted to plot their histograms to observe how the audio clips performed across the board.









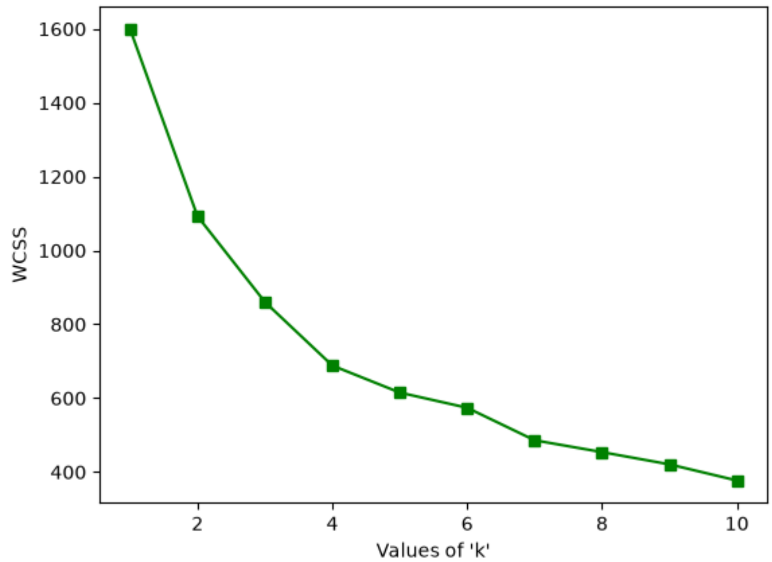
The histograms are generally right skewed with some anomalies especially in the spectral centroid histograms. While this could suggest that there are more animals that are softer or have lower variations of frequencies, it could also be the case that the softer animals are so soft to the point that they negate the loudness of the loud animals (or rather the loud animals are not loud enough).

Data Processing

Before delving into the clustering, I had to scale the features as they were fundamentally different in scale. This is important as if one feature has values that are exponentially greater in magnitude than that of other features, that clustering will be overwhelmingly biased to that feature.

Finding the optimal K value

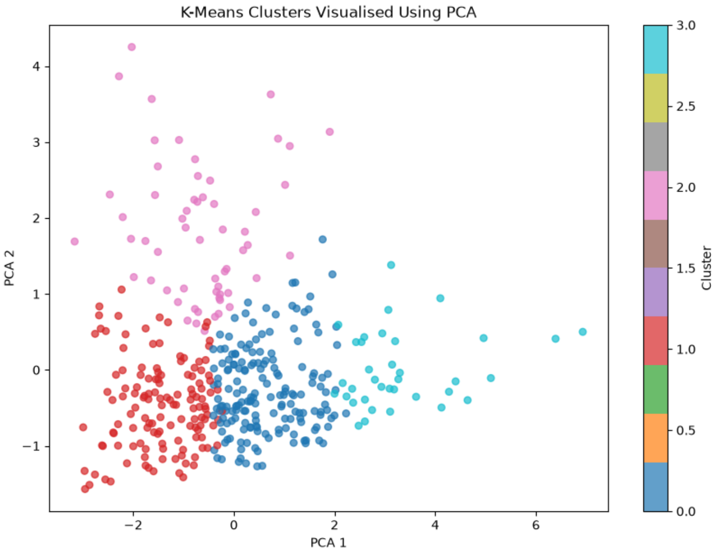
For K Means Clustering, the K value simply refers to the number of clusters the data should be grouped into. To deduce the optimal K value, I used the 'elbow' method where I choose a K value that is sufficiently large to fit the data well such that the next value K + 1 does not lead to a significant gain in clustering fit.



From this graph I chose 4 as there was a steep drop from 3 to 4 but the drop is gentle from 4 to 5. K = 5 is also a somewhat valid value of K here, depending on your definition of a 'gentle drop'.

Visualisation of Clustering using PCA

The number of features used for the clustering related to the number of 'dimensions' the data will be visualised in. It is non-trivial to visualise data in 4 dimensions so PCA is used to visualise data in 2 dimensions. It does so by using a weighted formula to illustrate the 4 features in 2 axes.



Certainly not the most pretty looking clustering but I was proud of 2 things- the points even if scattered stuck to their clustering region (clusters do not overlap) and 2 of the clusters (0 and 1) were compact, meaning that they shared similar acoustic features. However, to address the elephant in the room, the points in the remaining two clusters were very dispersed which suggests that the features chosen might not be the strongest discriminators.

That said, the PCA visualisation is not 100% reflective of the clusters that were formed in the K Means space. Thus, I also calculated the silhouette score of the clustering, which is essentially a measure of whether the points truly belong in their assigned clusters or if they would be a better fit elsewhere. The score came up to 0.319 which somewhat corroborates the PCA visualisation (i.e. some meaningful clustering but no strong separation based on the chosen features).

RMS ZCR Centroid Bandwidth

PC1 -0.160124 0.533575 0.617872 0.554881

PC2 0.975008 0.202123 0.091093 -0.014435

The PCA loadings revealed that PC2 was dominated by RMS and hence, loudness while PC1 was dominated by the spectral features.

Evaluation

There are many reasons why a clustering may not turn out perfect, from data related reasons to analysis related reasons.

Firstly, in my research, there were many features that I considered but did not experiment with (due to time constraints). One such feature includes MFCCs which gives valuable information regarding the frequency spectrum.

Secondly, in my feature extraction, I only considered taking the mean instead of say, the standard deviation. I think that is worth experimenting with as well since mean does not account for variation which is a key component of an audio clip.

Thirdly, the audio clips may not be perfect. If the microphone recording a dog was far compared to the microphone recording a cow, their loudness may not be too far off. Unfortunately, cleaning audio data is not as trivial as numbers as it requires the analyst to have a good ear and potentially some audio cleaning skills.

Acknowledgements

https://www.geeksforgeeks.org/machine-learning/kmeans-clustering-and-pca-on-wine-dataset/

https://medium.com/@rijuldahiya/a-comprehensive-guide-to-audio-processing-with-librosa-in-python-a49276387a4b

ChatGPT for vetting some code and breaking down musical concepts.